

Project Report

On

“ENGINEERING WEBSITE”

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In partial fulfilment for the award of

DIPLOMA

IN

COMPUTER ENGINEERING

Under the guidance of:

Prof. Geetanjali. S. Mahamunkar

at



Dr. Babasaheb Ambedkar Technological University, Lonere

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CERTIFICATE

This is to certify that the Project Report “Engineering Website” is the work carried out by SMARAN SANDEEP DAMBE(2211821), TEJAS PRAMOD CHOPADE(2211820), SWANIK SANTOSH GUDEKAR(2211835), MANNAS ABHIJEET BAKARE(2211821).

The students of final year diploma in Computer Engineering of Dr. Babasaheb Ambedkar Technological University Lonere during the academic year 2023-2024, in partial fulfilment of the requirements for the award of Diploma in Computer Engineering of Dr. Babasaheb Ambedkar Technological University (Diploma wing) Lonere.

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ACKNOWLEDGMENT

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Finally, would like to thank all those who helped us with the completion of this project.

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ABSTRACT

"Engineering Emporium" is the ultimate online destination catering to the diverse needs of engineers globally. Our platform offers a comprehensive range of essential products tailored to the engineering profession. From drafting tools and reference books to cutting-edge laptops, etc. we provide everything engineers need to thrive in their field. Our curated collection includes precision drafting instruments, drafting papers, and drawing aids, alongside a vast array of authoritative literature covering mechanical, electrical, civil, and software engineering disciplines. Additionally, we feature top-of-the-line laptops and computing devices optimized for engineering applications, ensuring peak performance and reliability. At "Engineering Emporium," we're committed to empowering engineers worldwide by providing access to high-quality products at competitive prices. Join us in fueling innovation and advancing the frontiers of engineering excellence.

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CHAPTER 1 : INTRODUCTION

1.1 Introductory Information

Welcome to DBATU Store, your one-stop destination for all the essential tools and equipment tailored specifically for engineers. Whether you're a student, a hobbyist, or a seasoned professional, we've got you covered with a wide range of products to suit your needs.

At DBATU Store, we understand the importance of having access to high-quality tools and materials to bring your engineering projects to life. That's why we've curated a comprehensive selection of products, ranging from electronic components and mechanical parts to software solutions and reference books.

Our mission is to make the process of acquiring engineering essentials as seamless and convenient as possible. With user-friendly navigation, secure payment options, and swift delivery services, we strive to provide you with an exceptional shopping experience from start to finish.

Whether you're working on a school assignment, a personal project, or a professional endeavor, you can trust DBATU Store to be your reliable partner in engineering excellence. Explore our catalog today and take the first step towards realizing your engineering aspirations!

DBATU Store is your premier destination for all things engineering. We offer an extensive range of products tailored to meet the needs of engineers from various disciplines. Our commitment to quality ensures that every item in our catalog meets the highest standards of craftsmanship and durability.

Shopping with us is easy and convenient, thanks to our user-friendly website and secure payment options. We prioritize fast shipping to ensure that you receive your orders promptly, wherever you are.

At DBATU Store, we're not just a retailer – we're a community of passionate engineers dedicated to innovation and collaboration. Follow us on social media to stay updated on the latest industry trends, product releases, and exclusive offers.

Your satisfaction is our top priority. If you have any questions or concerns, our responsive customer service team is here to help. Experience the difference with DBATU Store – where engineering meets excellence.

DBATU Store is more than just a marketplace; it's a hub for engineering enthusiasts and professionals alike. Our extensive product range covers all the essentials, from electronic components and mechanical parts to software solutions and reference materials. We pride ourselves on sourcing only the highest quality products from trusted manufacturers, ensuring that our customers have access to reliable tools and equipment.

One of our core values is accessibility. We believe that everyone should have the opportunity to pursue their engineering passions, which is why we offer competitive pricing without compromising on quality. Whether you're a student working on a project or a seasoned engineer tackling a complex problem, DBATU Store has everything you need to succeed.

But we're more than just a store. We're a community. Through our website and social media channels, we foster a space where engineers can connect, share ideas, and support one another. Our team is dedicated to providing exceptional customer service, ensuring that your shopping experience is smooth and enjoyable from start to finish.

At DBATU Store, we're committed to empowering the next generation of innovators and problem solvers. Join us on our mission to make engineering accessible to all.

1.2 SCOPE AND OBJECTIVE

1. Scope:

- **Product Range:** Offering a comprehensive selection of engineering essentials spanning various disciplines, including electronics, mechanics, software, and reference materials.
- **Customer Base:** Catering to engineers at all levels, from students and hobbyists to professionals and researchers.
- **Accessibility:** Ensuring that products are readily available and affordable, promoting inclusivity within the engineering community.
- **Community Engagement:** Providing platforms for engineers to connect, share knowledge, and collaborate through the store's website and social media channels.

2. Objectives:

- **Quality Assurance:** Ensuring that all products meet high-quality standards to meet the demands of engineering projects.
- **Customer Satisfaction:** Prioritizing customer experience by offering user-friendly interfaces, secure transactions, and responsive customer support.
- **Innovation:** Constantly updating the product offerings to incorporate the latest advancements and emerging technologies in the engineering field.
- **Education and Support:** Providing resources and guidance to help customers make informed decisions and excel in their engineering pursuits.
- **Community Building:** Fostering a vibrant community where engineers can connect, collaborate, and support each other in their professional and personal endeavors.
- **Sustainability:** Striving to minimize the environmental impact of operations and product offerings through responsible sourcing and eco-friendly practices.

By aligning its scope and objectives, DBATU Store aims to be more than just a marketplace; it aims to be a trusted partner and a hub of inspiration for engineers worldwide.

1.3FEATURES

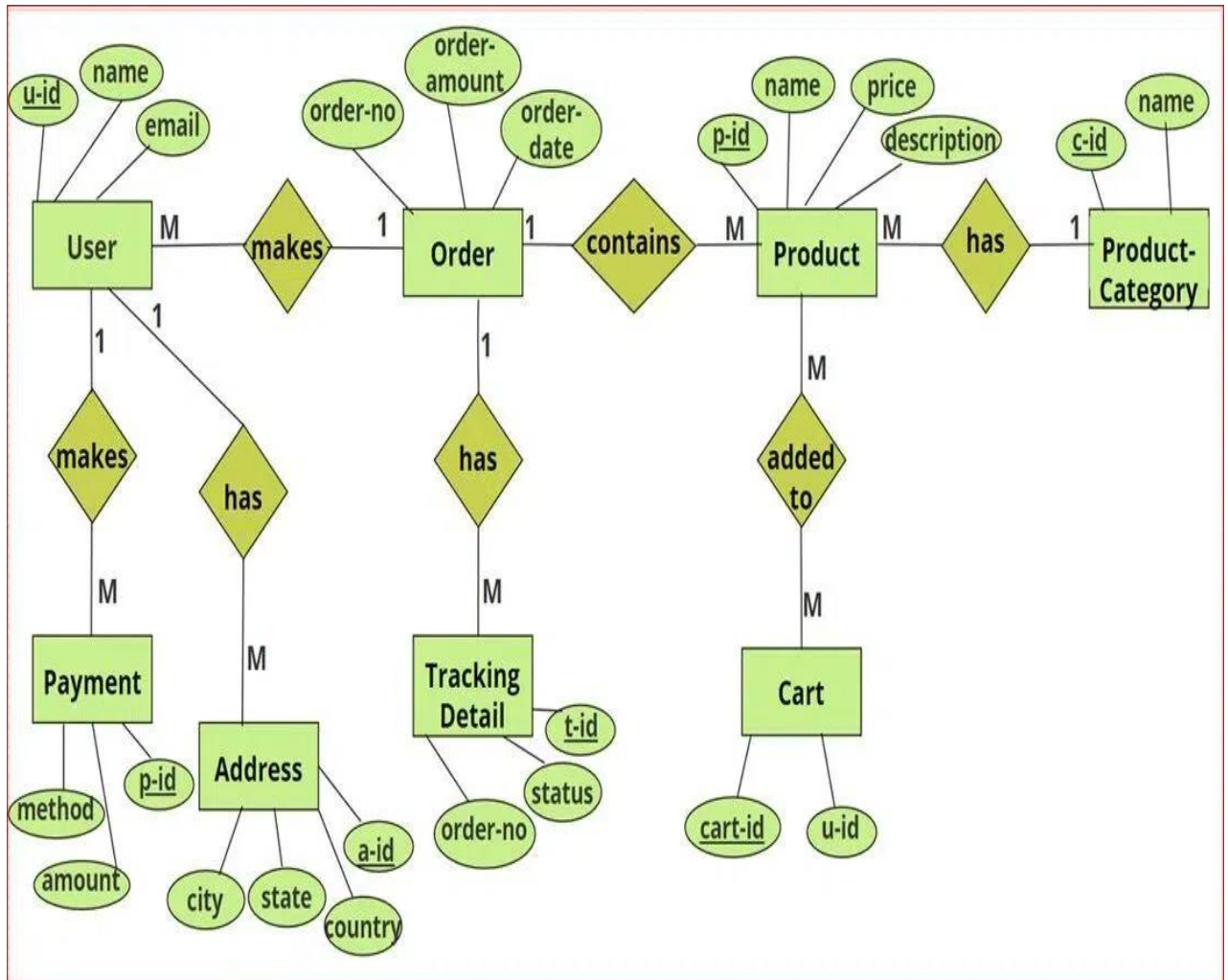
- **User-Friendly Interface:** The website is designed with a clean and intuitive interface, making it easy for customers to browse, search, and find the products they need. Filters and categories help streamline the shopping experience.
- **Secure Payment Options:** DBATU Store prioritizes the security of transactions. Customers can choose from various payment methods, including credit/debit cards, PayPal, and other secure payment gateways.
- **Fast and Reliable Shipping:** Orders are processed promptly, and customers can choose from different shipping options to suit their needs. Expedited shipping is available for those with urgent requirements.
- **Responsive Customer Support:** The store's customer support team is available to assist customers with any queries or concerns they may have. Whether it's product inquiries, order tracking, or technical assistance, support is just a message or call away.
- **Community Engagement:** DBATU Store fosters a sense of community among engineers through its blog, forums, and social media channels. Customers can connect with like-minded individuals, share insights, and seek advice on projects or career-related matters.
- **Regular Updates and Promotions:** Customers can stay informed about new product releases, special promotions, and discounts through email newsletters and social media announcements.
- **Educational Resources:** The store provides educational resources such as tutorials, guides, and recommended reading lists to help customers enhance their skills and knowledge in engineering.

- ❑ **Product Reviews and Ratings:** Customers can read reviews and ratings from fellow engineers to make informed purchasing decisions. They can also leave feedback on products they've purchased, contributing to the community's collective knowledge.
- ❑ **Responsive Design:** The website is optimized for various devices, including desktops, laptops, tablets, and smartphones, ensuring a seamless shopping experience regardless of the device used.
- ❑ **Bulk Ordering:** For business or educational institutions, DBATU Store offers bulk ordering options, allowing for efficient procurement of large quantities of products.
- ❑ **Customization Services:** Some products may be customizable or available for special order requests, catering to specific needs or project requirements.
- ❑ **Technical Support Resources:** In addition to customer service assistance, the store may provide technical support resources such as troubleshooting guides, FAQs, and video tutorials for complex products.
- ❑ **Reward Programs:** DBATU Store may offer loyalty programs or rewards for frequent shoppers, incentivizing repeat purchases and customer loyalty.
- ❑ **Integration with Engineering Software:** For software products, the store may offer seamless integration with popular engineering software platforms, enhancing workflow efficiency.

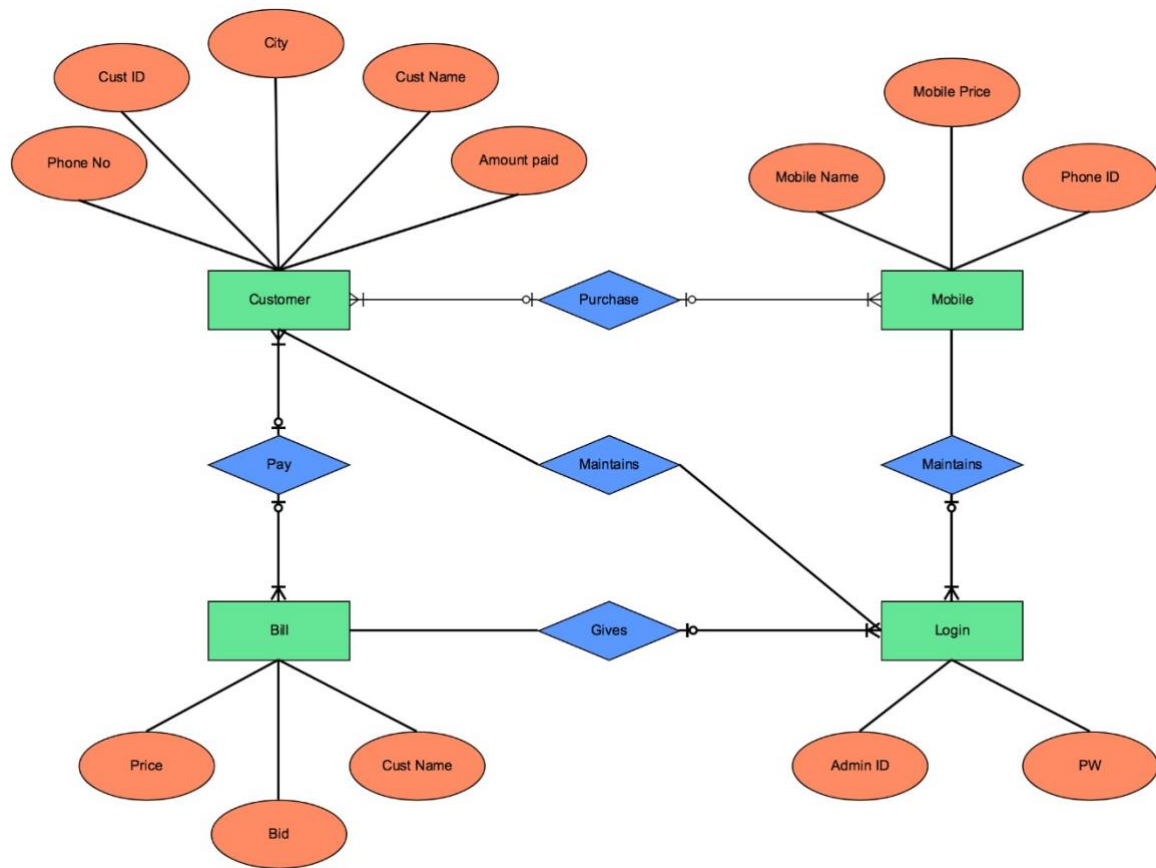


Chapter 2:

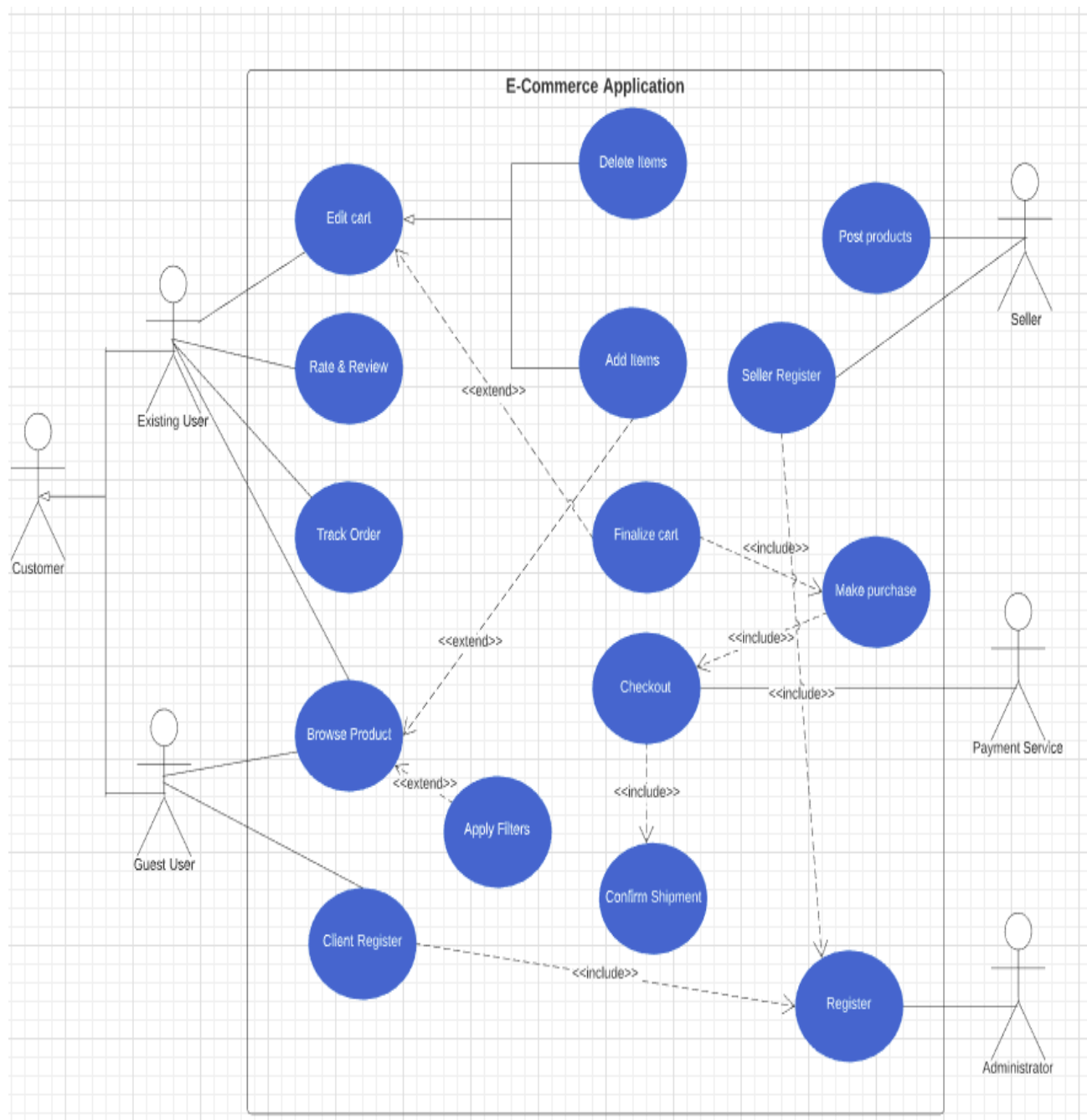
2.1 E-R diagram for DBATU STORE



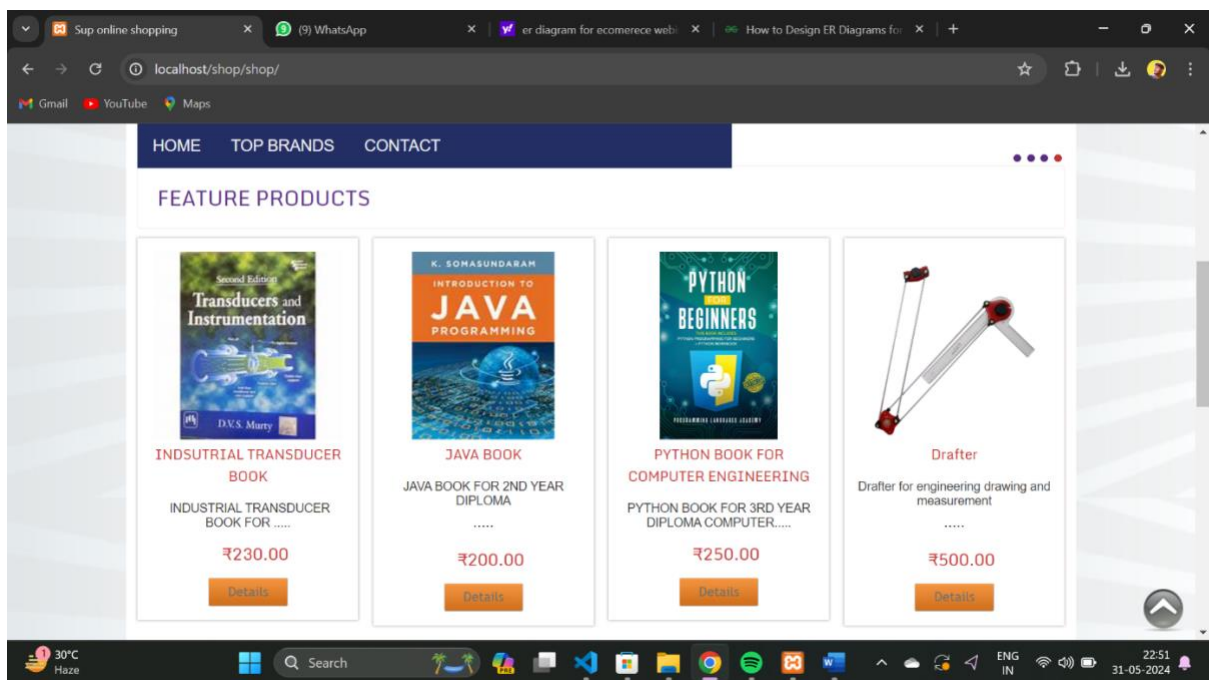
2.2 Admin panel:



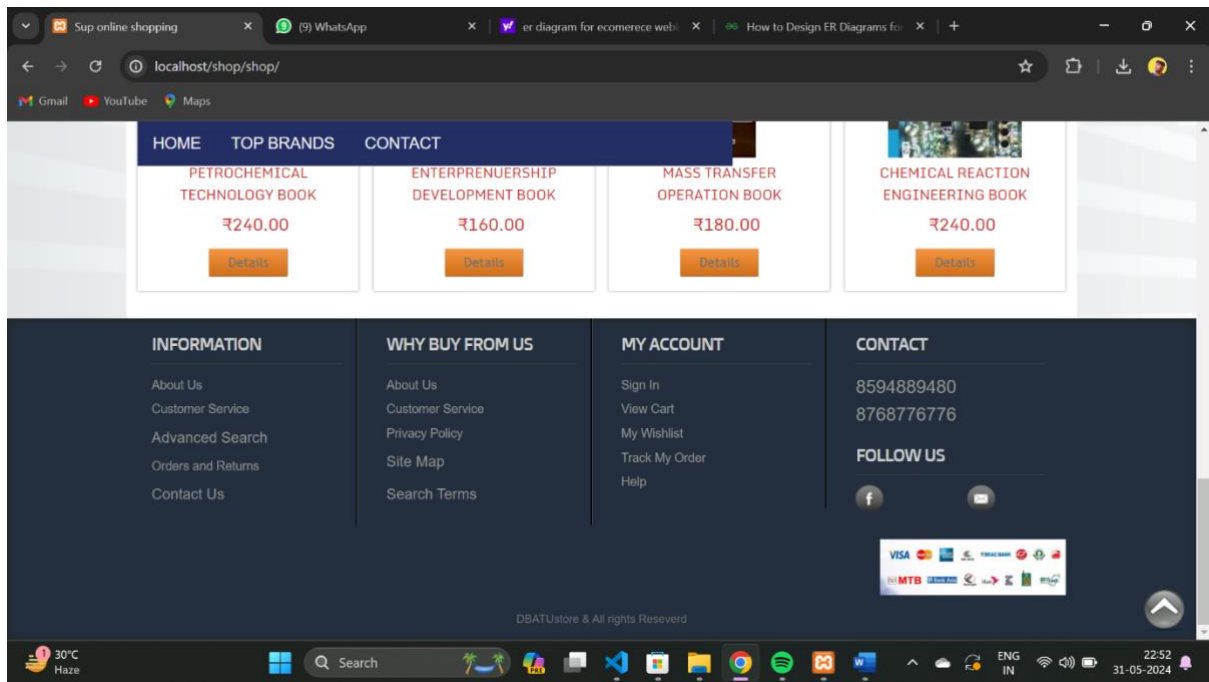
2.3 Usecase diagram



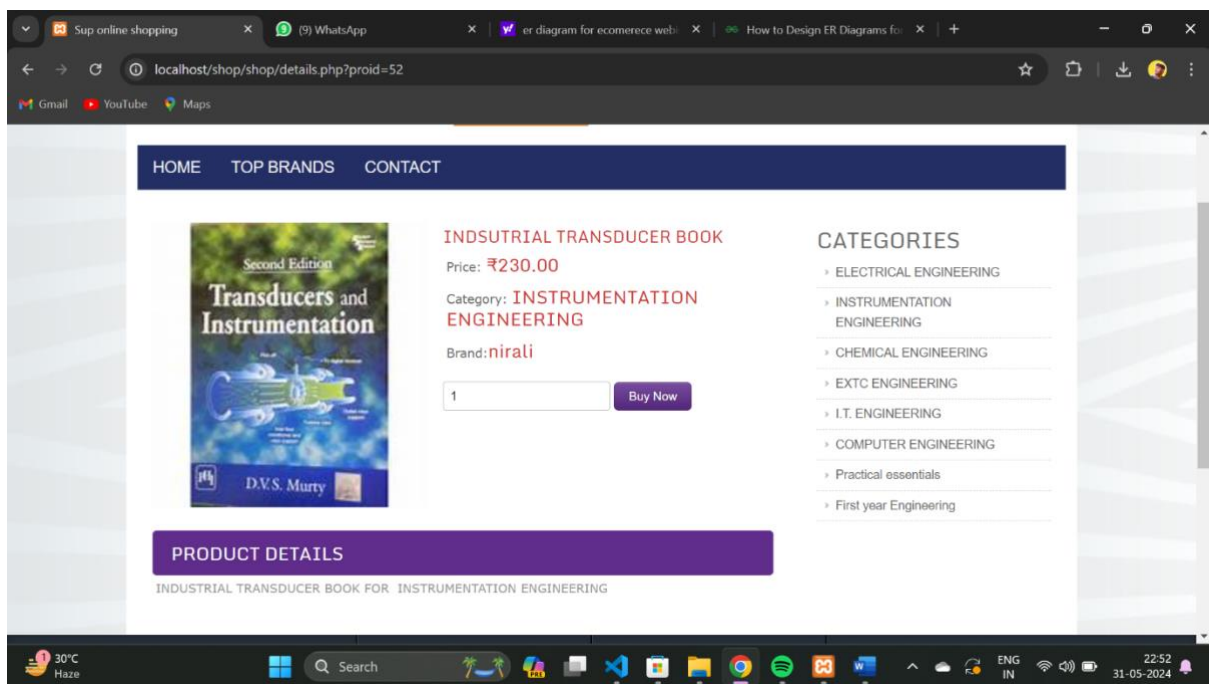
Chapter 3: Snapshots



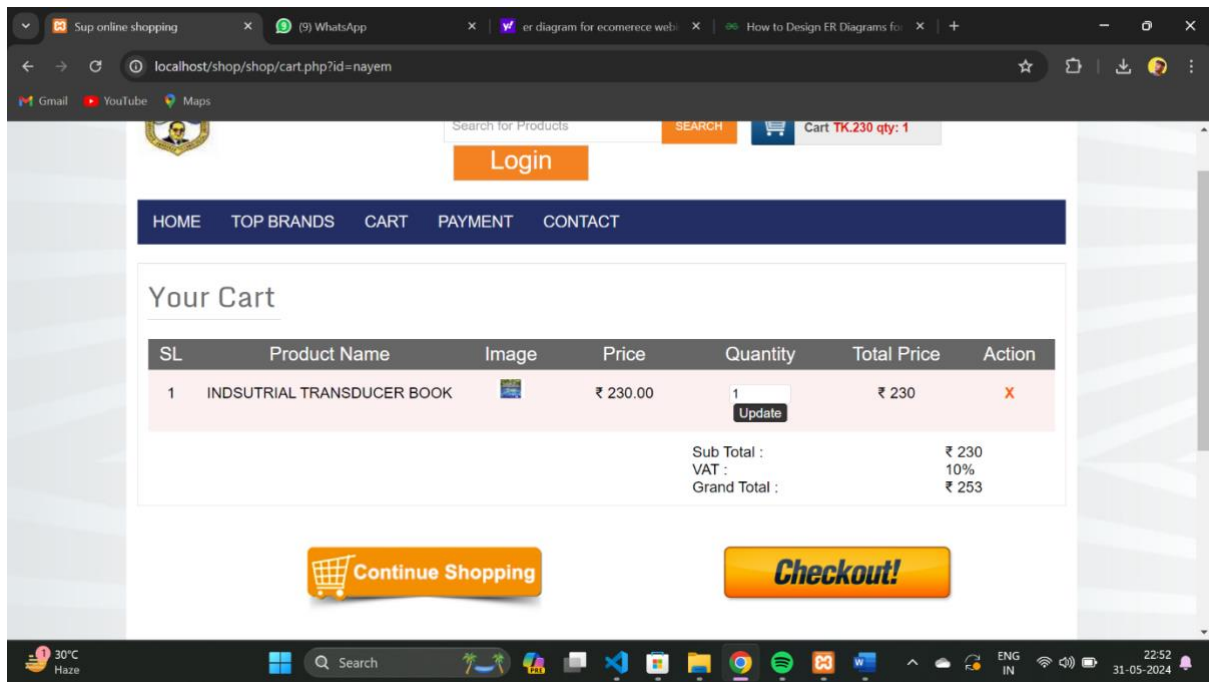
3.1 Above Snapshot of main page



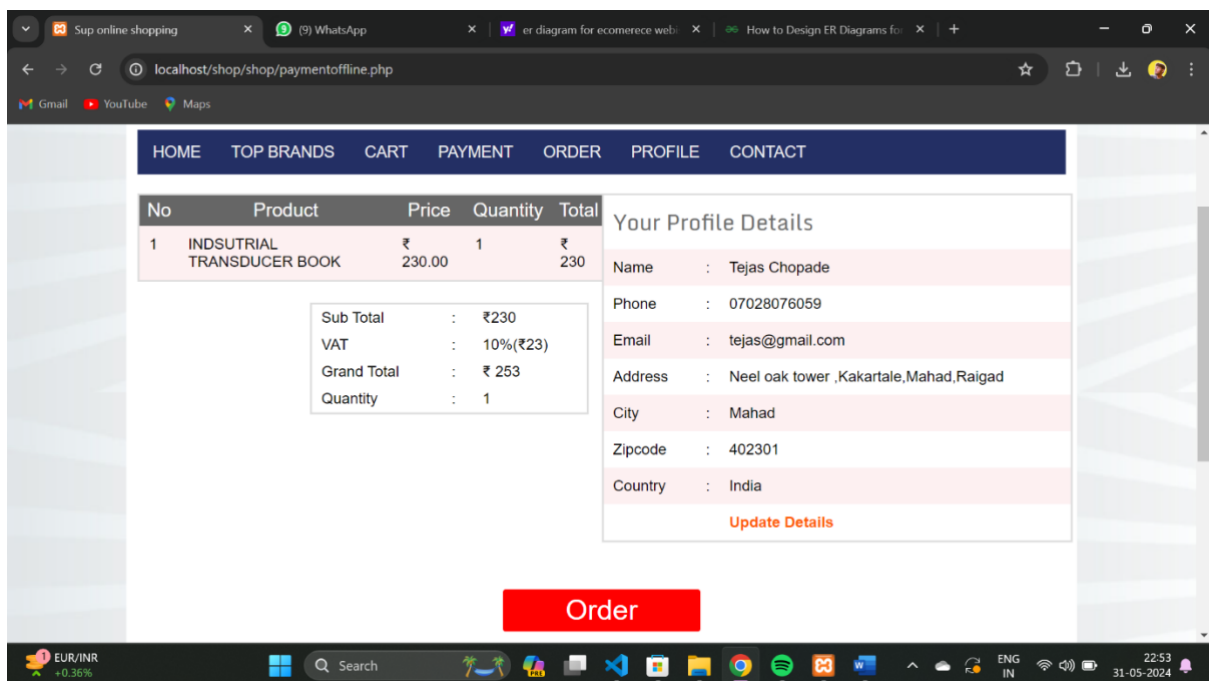
3.2 Above snapshot of footer section



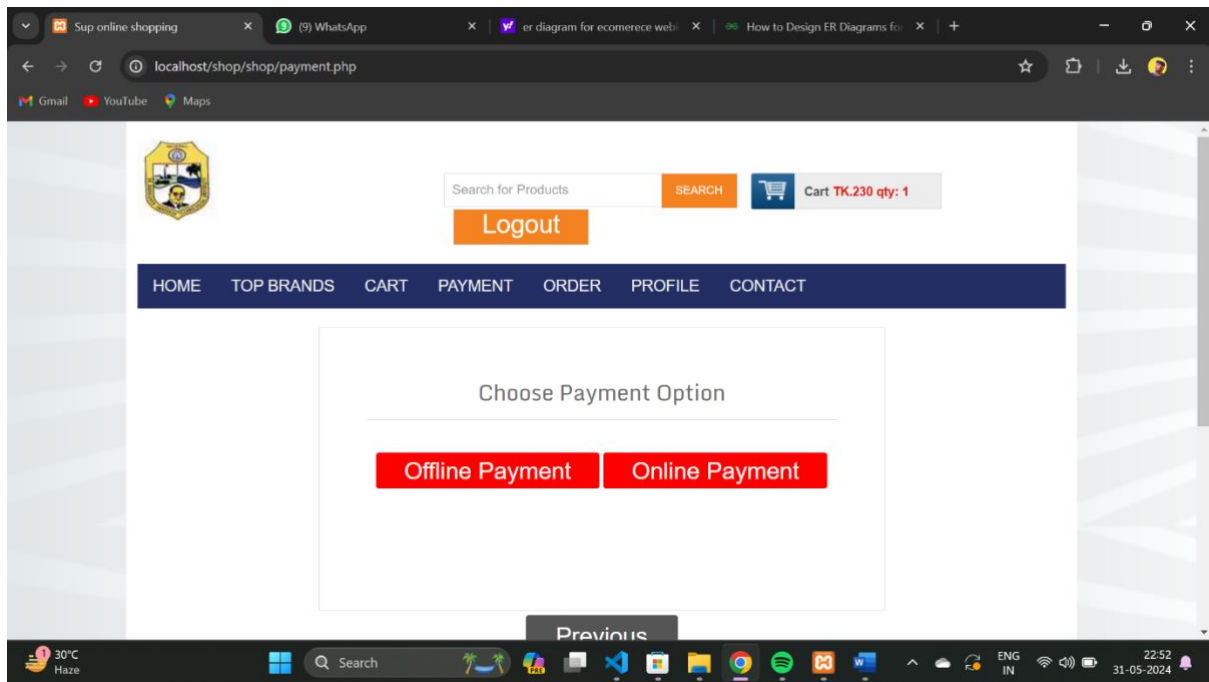
3.3 Above snapshot of product and its categories



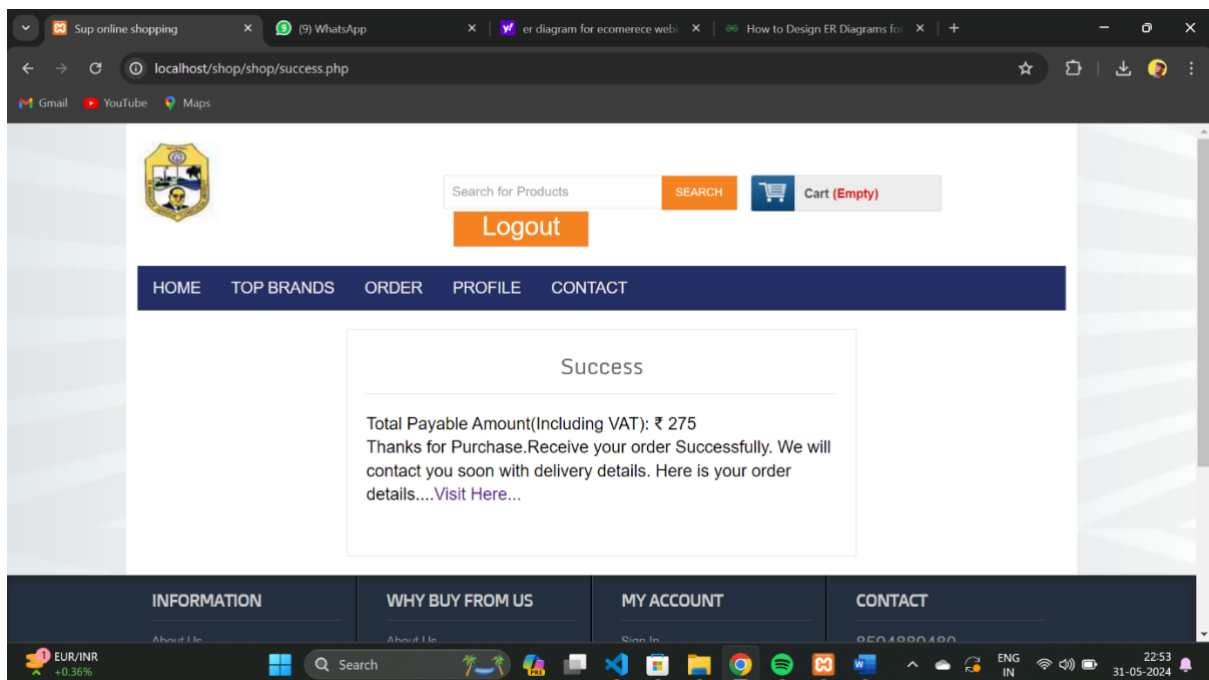
3.4 Above snapshot of product after clicking buy now button



3.5 Above snapshot is of order details of our product



3.6 Above snapshot is of payment option



3.7 Above snapshot is of successful order is placed in database

CHAPTER 4 : PROJECT IMPLEMENTATION

4.1 INTRODUCTION TO TECHNOLOGY

In this project we have used HTML, CSS, PHP, MY SQL and we have also used Xampp server in our project. Following is the introductory information about all :

1. HTML (HyperText Markup Language):

- HTML is the standard markup language used to create the structure and content of web pages.
- It consists of a series of elements that define the different parts of a webpage, such as headings, paragraphs, images, links, and forms.
- HTML uses tags to enclose and define the purpose of each element, making it readable by web browsers.

2. CSS (Cascading Style Sheets):

- CSS is a style sheet language used to control the presentation and layout of HTML documents.
- It allows developers to define styles for various elements, including fonts, colors, spacing, layout, and responsiveness.
- CSS enables the separation of content from presentation, making it easier to maintain and update the appearance of a website.

3. PHP (Hypertext Preprocessor):

- PHP is a server-side scripting language used for web development and dynamic content generation.
- It can be embedded within HTML to create dynamic web pages, interact with databases, process form data, and perform other server-side tasks.
- PHP is widely used for building web applications, content management systems (CMS), e-commerce platforms, and more.

4. MySQL:

- MySQL is an open-source relational database management system (RDBMS) that stores and manages structured data.
- It uses a structured query language (SQL) to interact with databases, allowing users to create, read, update, and delete data.

- MySQL is commonly used in web development for storing user information, product catalogs, transaction records, and other types of data.

5. XAMPP (Cross-Platform Apache, MySQL, PHP, and Perl):

- XAMPP is a free and open-source web server solution that provides a local development environment for web developers.
- It includes Apache (the web server), MySQL (the database), PHP (the server-side scripting language), and Perl (another scripting language) in a single package.
- XAMPP allows developers to set up a complete web development environment on their local machines, enabling them to develop, test, and debug web applications offline before deploying them to a live server.

Each of these technologies plays a crucial role in modern web development, empowering developers to create dynamic, interactive, and database-driven websites and web applications.

4.2 PROJECT IMPLEMENTATION

****Project Title:** DBATU Store - Online Engineering Essentials Marketplace**

****Project Description:****

DBATU Store aims to provide a convenient online platform for engineers to purchase a wide range of engineering essentials, including electronic components, mechanical parts, software tools, reference materials, and more. Users can create accounts, browse products, add items to their cart, and securely complete purchases.

****Technologies Used:****

1. HTML: For creating the structure and content of web pages.
2. CSS: For styling the web pages and enhancing the user interface.
3. PHP: For server-side scripting to handle user authentication, data processing, and interaction with the database.
4. MySQL: For storing product information, user details, and order records.
5. XAMPP: To set up a local development environment with Apache, MySQL, PHP, and other necessary components.

****Project Implementation Steps:****

1. ****Database Design****: Design the database schema to store product information, user details, and order records. Create tables for products, users, orders, and any other necessary entities.
2. ****Frontend Development****:
 - Create HTML templates for different pages such as home, product listings, user profile, shopping cart, and checkout.
 - Use CSS to style the HTML elements, design the layout, and enhance the user interface with colors, fonts, and responsive design.
3. ****Backend Development****:
 - Implement PHP scripts to handle user authentication (login, registration, logout).
 - Develop PHP scripts to interact with the MySQL database for tasks such as retrieving product listings, adding items to the shopping cart, and processing orders.
 - Ensure data validation and security measures to prevent vulnerabilities such as SQL injection and cross-site scripting (XSS).
4. ****Integration****:
 - Integrate the frontend HTML/CSS templates with the backend PHP scripts to create dynamic web pages.
 - Implement user authentication mechanisms to secure user accounts and protect sensitive information.
5. ****Testing****:
 - Test the functionality of the website by simulating user interactions such as browsing

products, adding items to the shopping cart, and completing orders.

- Perform cross-browser testing and ensure compatibility with popular web browsers.

6. **Deployment**:

- Deploy the project to a web server or hosting provider to make it accessible to users on the internet.
- Configure the server environment to support PHP and MySQL, and ensure proper security measures are in place.

7. **Maintenance and Updates**:

- Regularly update the website with new product listings, features, and bug fixes.
- Monitor server performance and security to ensure smooth operation and protect against potential threats.

In this way the overall project is implemented.

4.3 Technology used

Here's more brief information about the technology which is used:

1. **HTML (HyperText Markup Language)**:

- HTML5, the latest version of HTML, is likely used in the project to take advantage of its new features and improved semantics.
- Semantic HTML elements such as `<header>`, `<nav>`, `<main>`, `<section>`, and `<footer>` are employed to enhance accessibility and search engine optimization (SEO).
- HTML forms are utilized for user input, such as registration, login, search, and checkout forms.

2. **CSS (Cascading Style Sheets)**:

- CSS3 features like flexbox, grid layout, transitions, and animations may be utilized to create responsive and visually appealing layouts.
- Preprocessors like Sass or Less might be used to write CSS more efficiently, enabling features like variables, mixins, and nesting.
- CSS frameworks such as Bootstrap or Bulma could be utilized to speed up development and ensure consistency across the website's design.

3. **PHP (Hypertext Preprocessor)**:

- PHP is used to create dynamic web pages by embedding PHP code within HTML.
- Object-oriented programming (OOP) principles may be employed to organize PHP code into reusable classes and functions, enhancing maintainability and scalability.
- PHP frameworks like Laravel or Symfony might be utilized to expedite development, provide built-in security features, and enforce MVC (Model-View-Controller) architecture.

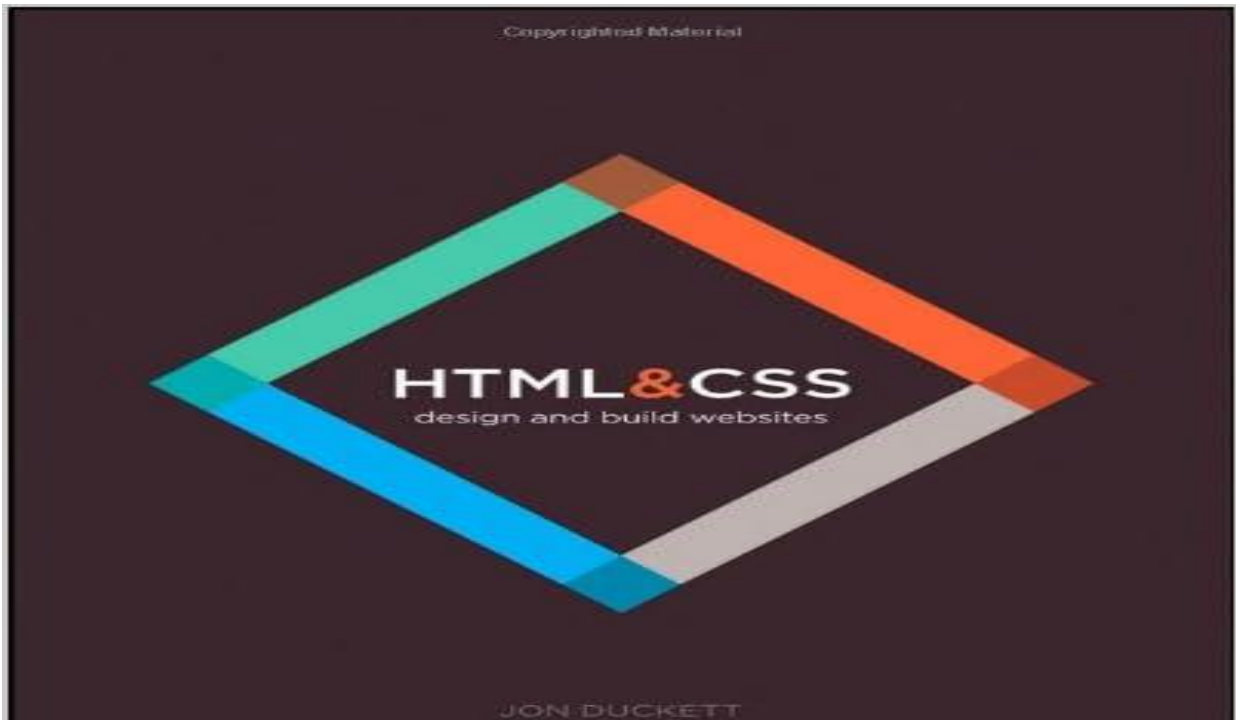
4. **MySQL**:

- MySQL is employed as the database management system to store structured data in tables with relationships between them.
- SQL queries are used to perform CRUD (Create, Read, Update, Delete) operations on the database, ensuring efficient data retrieval and manipulation.
- Indexing, normalization, and other optimization techniques are applied to improve database performance and scalability.

5. **XAMPP (Cross-Platform Apache, MySQL, PHP, and Perl)**:

- XAMPP provides a local development environment by bundling Apache HTTP Server, MySQL database, PHP interpreter, and other tools.
- Developers can use phpMyAdmin, included with XAMPP, to manage MySQL databases through a web interface.
- XAMPP simplifies the setup process for local development, allowing developers to test their code locally before deploying it to a live server.

These technologies collectively enable the creation of a robust, scalable, and secure online marketplace for engineering essentials, offering users a seamless shopping experience while empowering developers to build and maintain the platform effectively.



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CHAPTER 5 . ADVANTAGES AND DISADVANTAGES

5.1 ADVANTAGES

□ **Specialized Functionality:** By using HTML, CSS, PHP, MySQL, and XAMPP, developers can create specialized functionality tailored specifically to the needs of engineers. This includes

features such as product categorization by engineering discipline, advanced search filters for technical specifications, and integration with engineering software tools.

□ **Personalized User Experience:** With PHP and MySQL, developers can implement features for user account customization, such as saved preferences, order history, and personalized recommendations based on past purchases or browsing behavior. This enhances the user experience and fosters customer loyalty.

□ **Real-Time Inventory Management:** PHP and MySQL enable real-time inventory management, allowing the DBATU Store to track product availability, update stock levels, and prevent overselling. This ensures that customers have accurate information about product availability and reduces the likelihood of order cancellations due to stockouts.

□ **Secure Payment Processing:** PHP and MySQL facilitate secure payment processing by integrating with payment gateways and implementing encryption protocols to protect sensitive financial information. This instills trust and confidence in customers, leading to higher conversion rates and increased sales for the DBATU Store.

□ **Analytics and Reporting:** PHP and MySQL enable the implementation of analytics and reporting tools to track key performance indicators (KPIs) such as website traffic, conversion rates, and customer demographics. This valuable data can be used to make informed business decisions, optimize marketing strategies, and improve the overall performance of the DBATU Store.

□ **Multi-Language and Multi-Currency Support:** With PHP and MySQL, developers can implement multi-language and multi-currency support, allowing the DBATU Store to cater to a global audience. This expands the market reach of the store and increases opportunities for sales and growth.

□ **Mobile Responsiveness:** HTML and CSS support mobile responsiveness, enabling the DBATU Store to provide a seamless browsing and shopping experience across devices of all sizes, including smartphones and tablets. This accessibility enhances user engagement and satisfaction, leading to higher retention rates and increased sales.

□ **Integration with Third-Party Services:** PHP and MySQL facilitate integration with third-party services such as shipping carriers, inventory management systems, and customer relationship management (CRM) software. This streamlines business operations, reduces



5.2 DISADVANTAGES

1. **Learning Curve:** Developing and maintaining a project using HTML, CSS, PHP, MySQL, and XAMPP requires proficiency in multiple technologies. This could pose a challenge for developers who are not familiar with all aspects of the technology stack.

2. **Security Risks:** PHP and MySQL are susceptible to security vulnerabilities if not properly configured and maintained. Poorly written PHP code or inadequate security measures can expose the DBATU Store to risks such as SQL injection, cross-site scripting (XSS), and data breaches.
3. **Scalability Issues:** While HTML, CSS, PHP, and MySQL offer scalability, poorly optimized code or database design could lead to performance bottlenecks as the DBATU Store grows. Scaling the project to accommodate increased traffic and data volume may require significant architectural changes and resource allocation.
4. **Dependency on Third-Party Services:** Integrating with third-party services for payment processing, shipping, and other functionalities introduces dependencies that are beyond the project's control. Any disruptions or changes to these services could impact the functionality and user experience of the DBATU Store.
5. **Maintenance Overhead:** Regular maintenance and updates are required to ensure the security, performance, and functionality of the DBATU Store. This includes patching vulnerabilities, optimizing code and database queries, and implementing new features based on user feedback and market trends.
6. **Cost of Infrastructure:** While HTML, CSS, PHP, MySQL, and XAMPP are open-source technologies, hosting the DBATU Store on a reliable server and ensuring adequate resources for performance and scalability may incur ongoing costs.
7. **Competitive Landscape:** The online marketplace for engineering essentials is highly competitive, with established players and niche competitors vying for market share. Differentiating the DBATU Store and attracting customers in this crowded space may require significant marketing efforts and strategic initiatives.

8. **Regulatory Compliance:** Depending on the geographic location and target market of the DBATU Store, it may need to comply with various regulations and standards related to data privacy, consumer protection, and e-commerce transactions. Ensuring compliance adds complexity and potential legal liabilities to the project.

By carefully addressing these potential disadvantages and implementing strategies to mitigate risks, developers can maximize the success and sustainability of the DBATU Store project while delivering value to its users and stakeholders.

CHAPTER 6 . CONCLUSION

In conclusion, the DBATU Store project stands as a promising endeavor to provide engineers with a comprehensive and user-friendly platform for their essential needs. Leveraging the robust technologies of HTML, CSS, PHP, MySQL, and XAMPP, it offers a host of advantages including specialized functionality, personalized user experiences, real-time inventory management, secure payment processing, analytics capabilities, and more.

While the project presents numerous opportunities, it's imperative to address potential challenges such as the learning curve, security risks, scalability concerns, and regulatory compliance. By proactively mitigating these risks and continuously improving the platform, the DBATU Store can position itself as a reliable and indispensable resource for engineers of all backgrounds and specialties.

The DBATU Store project represents a pivotal initiative aimed at revolutionizing the way engineers access and procure essential tools, resources, and equipment. Through the adept utilization of HTML, CSS, PHP, MySQL, and XAMPP, this platform offers a multitude of advantages, including specialized functionality, personalized user experiences, real-time inventory management, secure payment processing, analytics capabilities, and seamless integration with third-party services.

Despite the project's many strengths, it is essential to recognize and address potential challenges such as the learning curve associated with the technology stack, security vulnerabilities, scalability concerns, dependency on third-party services, maintenance overhead, infrastructure costs, competitive landscape, and regulatory compliance. By proactively mitigating these risks and embracing a strategy of continuous improvement, the DBATU Store can cement its position as a leading online marketplace for engineering essentials.

With a commitment to excellence, innovation, and customer satisfaction, the DBATU Store has the potential to not only meet but exceed the expectations of its users, fostering a vibrant community and contributing to the advancement of engineering endeavors worldwide.

Ultimately, the success of the DBATU Store hinges on its ability to deliver tangible value to engineers of all disciplines and levels of expertise. With a relentless focus on innovation, user experience, and community engagement, the DBATU Store has the potential to transcend boundaries, foster collaboration, and empower engineers to achieve their goals and aspirations. As the project evolves and matures, it is poised to make a significant impact on the engineering landscape, driving progress and innovation in the years to come.



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CHAPTER 7 : REFERENCE

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